

MILESTONE HEALTH REVIEW

Milestone 1 Health Review Report

EMBRAPII Reports RAG — Project Scaffold & Docker

PROJECT

EMBRAPII Reports RAG

REVIEW DATE

2026-06-30

BRANCH REVIEWED

`milestone-01e-health-script-readme`

MILESTONE

1 — Project Scaffold & Docker

REVIEWER

Cursor agent

OVERALL STATUS

Complete — no blockers

MILESTONE STATUS

Complete

For intended scope

BLOCKING ISSUES

0

None found

ACCEPTANCE CRITERIA

13 / 13

All passed

HARDENING ITEMS

9

3 medium · 6 low

Executive Summary

Milestone 1 is complete for its intended scope. The repository now has a local Docker-based scaffold with five services: **frontend**, **backend**, **worker**, **Redis**, and **Qdrant**. Static validation passed after installing backend development dependencies, and runtime validation confirmed that the full Docker stack builds, starts, and passes the health script.

The project is moving in the right direction. It matches the MVP stack described in

`PROJECT_DECISIONS.md`, keeps RAG behavior out of scope for Milestone 1, documents current limitations honestly, and leaves a clean baseline for Milestone 2.

Bottom line: No blocker or high-severity issue was found. The main recommendations before or during Milestone 2 are to add Docker Compose healthchecks, make backend dependency health more explicit once real Redis and Qdrant clients are introduced, pin the Qdrant image instead of using `latest`, add Prettier configuration, and consider CI for repeatable validation.

Scope & Review Criteria

The review used these source-of-truth documents:

- `PROJECT_DECISIONS.md`
- `docs/specs/milestone-01-overview.md`
- `docs/specs/milestone-01a-root-docker-compose.md`
- `docs/specs/milestone-01b-backend-health.md`
- `docs/specs/milestone-01c-worker-health.md`
- `docs/specs/milestone-01d-frontend-shell.md`
- `docs/specs/milestone-01e-health-script-readme.md`
- `README.md`

Milestone 1 was expected to deliver

- Root Docker Compose setup for frontend, backend, worker, Redis, and Qdrant.
- Minimal FastAPI backend with `/health`.
- Minimal worker container or process with visible status behavior.
- Minimal React + TypeScript + Vite frontend shell.
- Persistent local Qdrant storage under `data/qdrant/`.
- Local document directory under `data/documents/`.
- Host health script for backend, Qdrant, Redis, worker, and optional frontend.
- README instructions and non-secret `.env.example`.

Out of scope for this milestone

PDF ingestion, chunking, embeddings, indexing, retrieval, Q&A, summaries, comparisons, evidence audit, golden-set evaluation, LLM calls, API key handling, and Redis + RQ background jobs.

What Was Completed

Root Scaffold

- `docker-compose.yml` defines five services: `frontend`, `backend`, `worker`, `redis`, and `qdrant`.
- Default ports match the project decisions: Frontend `5173`, Backend `8000`, Qdrant `6333`, Redis `6379`.
- `data/documents/` and `data/qdrant/` exist for local document and vector-store persistence.
- `.env.example` contains non-secret local defaults.

- `.gitignore` excludes `.env`, local databases, generated frontend output, dependency folders, and runtime Qdrant/document contents while preserving `.gitkeep` placeholders.

Backend Scaffold

- `backend/pyproject.toml` defines the FastAPI backend with `uv`, `pytest`, and `ruff`.
- `backend/app/main.py` creates the FastAPI app and registers the health router.
- `backend/app/api/routes/health.py` exposes `GET /health`.
- `backend/app/api/schemas/health.py` defines the response shape.
- `backend/app/core/config.py` and `backend/app/core/dependencies.py` provide the initial configuration and dependency injection pattern.
- Backend tests cover the health endpoint response shape.

Worker Scaffold

- `backend/app/worker/main.py` implements a placeholder long-running worker process.
- The worker reports that it is a Milestone 1 placeholder and that real RQ background jobs are planned for a later milestone.
- Worker tests cover importability, status construction, and startup logging.

Frontend Scaffold

- `frontend/package.json` defines a React + TypeScript + Vite app with `build`, `lint`, `dev`, and `preview` scripts.
- `frontend/src/App.tsx` displays a Portuguese MVP shell that clearly says upload, Q&A, summary, comparison, and evidence audit are not functional yet.
- The frontend presents planned capabilities without faking completed RAG features.

Health & Documentation

- `scripts/check-health.sh` verifies: Backend `/health`, Qdrant `/healthz`, Redis `PING`, Worker container running, and Frontend availability as an optional check.
- `README.md` documents quick start, stack ports, health checks, environment variables, what works, and what is not implemented yet.

Acceptance Criteria Results

CRITERION	RESULT	EVIDENCE
<code>docker compose up --build</code> starts all five services	Pass	Detached Docker startup completed successfully and all services were listed as running.
Frontend reachable at <code>http://localhost:5173</code>	Pass	HTTP status probe returned <code>200</code> ; health script also passed optional frontend check.

CRITERION	RESULT	EVIDENCE
Backend reachable at <code>/health</code> with <code>status: ok</code>	Pass	Backend returned <code>{"status":"ok","service":"backend","version":"0.1.0"...}</code> .
Qdrant reachable on port <code>6333</code>	Pass	<code>curl http://localhost:6333/healthz</code> returned <code>healthz check passed</code> .
Redis responds to <code>PING</code>	Pass	<code>redis-cli ping</code> returned <code>PONG</code> .
Worker container starts and is distinct from backend	Pass	<code>docker compose ps</code> showed <code>worker</code> as a separate running service; logs showed placeholder worker startup.
<code>scripts/check-health.sh</code> reports all required services healthy	Pass	Health script exited <code>0</code> and reported all required checks as passed.
Backend tests pass	Pass	<code>6 passed, 1 warning in 0.41s</code> .
Backend lint and format checks pass	Pass	Ruff reported <code>All checks passed!</code> and <code>14 files already formatted</code> .
Frontend build passes	Pass	TypeScript and Vite production build completed successfully.
Frontend lint passes	Pass	ESLint completed with exit code <code>0</code> .
Docker Compose config validates	Pass	<code>docker compose config</code> rendered successfully.
README instructions match actual commands	Pass	Documented commands match the working validation path.

Validation Evidence

Static Checks

COMMAND	RESULT	NOTES
<code>docker compose config</code>	Pass	Compose rendered five expected services.
<code>bash -n scripts/check-health.sh</code>	Pass	Bash syntax valid.
<code>uv --directory backend sync</code>	Pass	Backend dependencies resolved and installed.

COMMAND	RESULT	NOTES
<code>uv --directory backend run pytest -q</code>	Pass	6 tests passed, 1 Starlette/TestClient deprecation warning.
<code>ruff check .</code> (temp cache dir)	Pass	Used temp cache because existing <code>.ruff_cache</code> had permission issues.
<code>ruff format --check .</code> (temp cache dir)	Pass	14 files already formatted.
<code>npm --prefix frontend run build</code>	Pass	Build completed; npm emitted a local warning about unknown env config <code>devdir</code> .
<code>npm --prefix frontend run lint</code>	Pass	ESLint completed successfully; same npm <code>devdir</code> warning appeared.

Runtime Checks

COMMAND	RESULT	EVIDENCE
<code>docker compose up --build -d</code>	Pass	Backend, frontend, worker images built; containers started.
<code>bash scripts/check-health.sh</code>	Pass	All required Milestone 1 checks passed.
<code>docker compose ps</code>	Pass	All five services were running.
<code>curl -fsS http://localhost:8000/health</code>	Pass	Backend returned status <code>ok</code> .
<code>curl -fsS http://localhost:6333/healthz</code>	Pass	Qdrant returned <code>healthz check passed</code> .
<code>redis-cli ping</code>	Pass	Redis returned <code>PONG</code> .
<code>docker compose logs worker --tail=30</code>	Pass	Logs showed placeholder worker startup and configured Redis/Qdrant URLs.
<code>curl ... http://localhost:5173</code>	Pass	Frontend returned <code>200</code> .

Note on first-pass failures: Some early commands failed due to local environment invocation issues, not project defects — `uv run pytest` / `ruff` ran before `uv sync`, initial parallel npm commands resolved from the repository root, and sandboxed `curl localhost` calls could not reach host ports. These were corrected during validation and are *not* classified as Milestone 1 implementation failures.

Architecture & Direction Assessment

The project is aligned with the intended MVP direction.

Positive signs

- The Docker stack matches the planned local MVP architecture.
- The service names, ports, and local data paths match `PROJECT_DECISIONS.md`.
- The backend starts with thin routing, typed response schemas, configuration, and dependency injection structure.
- The worker exists as a separate process, preparing for later Redis + RQ jobs without pretending they exist now.
- The frontend shell is honest about the current state and communicates planned capabilities in Portuguese.
- The README clearly separates what works in Milestone 1 from later RAG workflows.
- No hardcoded LLM provider keys or secrets were found.
- RAG features are not faked — upload, ingestion, embeddings, retrieval, Q&A, summaries, comparisons, evidence audit, and evaluation remain correctly deferred.

Expected deferrals

- Full pragmatic hexagonal backend layout is not present yet — acceptable because `PROJECT_DECISIONS.md` assigns the backend architecture skeleton to Milestone 2.
- Redis + RQ job implementation is not present yet — acceptable because Milestone 1 only required a distinguishable worker scaffold.
- `docs/evaluation/` and `docs/adr/` are not present yet — acceptable until evaluation work or new architecture decisions begin.

Findings



Medium Severity

Docker Compose does not define service-level `healthcheck:` blocks **MEDIUM**

Impact: `depends_on` currently means `service_started`, not `service_healthy`. On cold starts, backend or worker may start before Redis or Qdrant are actually ready.

Recommendation: Add Compose healthchecks for backend, Redis, Qdrant, and possibly frontend during hardening or before Milestone 2 runtime workflows depend on them.

Backend `/health` does not live-probe Redis or Qdrant **MEDIUM**

Impact: The backend can return `status: ok` while dependency health is reported as `not_configured`, even when URLs are configured.

Recommendation: Keep acceptable for Milestone 1, but update health behavior when real Redis and Qdrant clients are introduced.

Qdrant uses `qdrant/qdrant:latest` **MEDIUM**

Impact: Reproducibility risk. A future image update could change behavior or health endpoints.

Recommendation: Pin Qdrant to a tested version.

Low Severity

Worker health is based on container running state only **LOW**

Impact: Acceptable for a placeholder worker, but not enough once real background jobs exist.

Recommendation: Add a worker heartbeat, queue connectivity check, or job system health signal in the Background Jobs milestone.

Frontend uses Vite dev server in Docker **LOW**

Impact: Acceptable for the local MVP scaffold, but less production-like than a static build served by nginx or Vite preview.

Recommendation: Revisit when moving from local MVP to deployment-oriented packaging.

Prettier is not configured **LOW**

Impact: `PROJECT_DECISIONS.md` expects ESLint and Prettier for frontend consistency. ESLint exists, but Prettier does not.

Recommendation: Add Prettier config and scripts during frontend hardening.

No CI pipeline exists **LOW**

Impact: Validation is currently manual/local.

Recommendation: Add CI once the project has a stable integration branch and the first few milestones are merged.

Existing `.ruff_cache` permission issue affected first-pass validation **LOW**

Impact: Ruff itself passes with a temp cache, but local cache ownership could confuse future agents.

Recommendation: Clean or recreate `.ruff_cache` outside this review if it continues to appear.

npm emits an unknown `devdir` config warning **LOW**

Impact: Not a build or lint failure. It appears to come from the local npm environment, not this repository.

Recommendation: Ignore for Milestone 1 unless it starts affecting npm commands.

Conclusions

Milestone 1 achieved its purpose: it established a runnable local foundation for the EMBRAPII Reports RAG MVP.

The scaffold is ready to support Milestone 2, especially the backend architecture skeleton. The most important next step is to preserve the current scope discipline: add real architecture boundaries and contracts before implementing ingestion or RAG workflows.

No implementation problem found in this review should block progress. The medium-severity findings are hardening items, not acceptance failures.

Recommended Next Actions

- 1 Proceed to Milestone 2: Backend Architecture Skeleton.
- 2 Add the pragmatic hexagonal folder structure before implementing ingestion logic.
- 3 Keep FastAPI routes thin and move workflow coordination into application services or use cases.
- 4 Add ports only where replacement is realistic: parser, embedding provider, retrievers, repositories, file storage, and LLM provider factory.
- 5 Consider adding Compose healthchecks before starting Milestone 3, because ingestion and retrieval will depend on Qdrant and Redis readiness.
- 6 Pin Qdrant to a tested version.
- 7 Add Prettier configuration when frontend work becomes more active.
- 8 Preserve README honesty by updating it whenever workflows move from planned to functional.

Appendix A — Important Runtime Outputs

```
BACKEND HEALTH{"status":"ok","service":"backend","version":"0.1.0","dependencies":
{"redis":"not_configured","qdrant":"not_configured"}}
```

HEALTH SCRIPTChecking Milestone 1 stack health from /home/kaique/embrapii_rag

```
[PASS] Backend (http://localhost:8000/health)
[PASS] Qdrant (http://localhost:6333/healthz)
[PASS] Redis (PING)
[PASS] Worker container (worker)
[PASS] Frontend (http://localhost:5173) [optional]
```

All required Milestone 1 health checks passed.

```
DOCKER SERVICESembrapii_rag-backend-1    Up    0.0.0.0:8000->8000/tcp
embrapii_rag-frontend-1    Up    0.0.0.0:5173->5173/tcp
embrapii_rag-qdrant-1      Up    0.0.0.0:6333->6333/tcp
embrapii_rag-redis-1       Up    0.0.0.0:6379->6379/tcp
embrapii_rag-worker-1      Up    8000/tcp
```

WORKER STARTUPWorker service: worker

Worker status: placeholder (Milestone 1 placeholder)

App version: 0.1.0

Redis URL configured: yes

Qdrant URL configured: yes

Milestone 1 placeholder worker started. Background jobs (RQ) will be implemented in a later milestone.

BACKEND TESTS6 passed, 1 warning in 0.41s

```
FRONTEND BUILDvite v6.4.3 building for
production...
29 modules transformed.
built in 923ms
```

Appendix B — Report Generation

- Markdown source: `docs/reports/milestone-01-health-report.md`
- PDF export: `docs/reports/milestone-01-health-report.pdf`

No project runtime dependency was added for PDF generation.